20-EECE-5180C/6080C	**Introduction to VLSI Design**
Required/Elective:	Elective for EE and CE Majors and Required for VLSI Systems Minors.
Catalog Data:	20-EECE-5180C/6080C. Introduction to VLSI Design. Credits 4. Introduces techniques and tools for scalable VLSI design and analysis. Emphasis is on circuit and physical design and on performance analysis. Includes lab experiments with hands-on usage of CAD tools.
Prereqs by Course/Topic:	1. Logic Design - combinational and sequential. (EECE2060) 2. Basic Network Analysis (EECE2050C) 3. Electronics - MOS transistor theory. (EECE3071) 4. Computer organization and microprocessors. (EECE3026) 5180C – Senior status is required 6080C – Graduate status is required. Graduate students are expected to have mastered the above topics.
Textbook:	Neil Weste and D. Harris, "Principles of CMOS VLSI Design : A Circuits and Systems Perspective", Addison-Wesley (latest edition).
References:	Jan M. Rabaey, "Digital Integrated Circuits: A Design Perspective," Prentice Hall (latest edition).
Goals:	Upon completion of this course, the students will be able to develop, layout and simulate fabricatable designs of VLSI building-block modules, such as registers and arithmetic units, and medium-scale VLSI chips.
Topics:	1. Basic MOS Transistor Theory 2. Circuit Simulation, Timing Analysis 3. Performance Estimation, resistance, capacitance, power, fall and rise times. 4. Transistor Sizing 5. Scalable Designs and Design Rules 6. Charge Sharing 7. CMOS logic families and circuit techniques. 8. Latch-Up and solutions. 9. Clocks and multiphase clocking. 10. VLSI Building Blocks - Inverters, Gates, Multiplexers, Registers, Arithmetic and Logic Units. 11. Controller design. 12. Hierarchical, modular and bit-sliced design techniques. 13. Pipelining 14. Area-Time-Power Tradeoffs in VLSI Design 15. Design with standard cells. 17. Clock distribution (tentative)
Class/Laboratory Schedule:	class meets 2 times a week for 90 minutes each time; Lab component is integrated into the course.
Additional Work for Graduate Students	6080C graduate students will be required to demonstrate in-depth understanding by meeting specific homework, project and/or exam requirements designated for them.
Course Learning Objectives:	Students will, 1. Be able to design, across logic, circuit and layout levels of abstraction, a small VLSI microchip in CMOS. 2. Be able to use CAD tools for VLSI circuit and physical design.
Outcomes:	1. (c) an ability to design a system, component, or process to meet desired needs within realistic constraints. 2. (e) an ability to identify, formulate, and solve engineering problems. 3. (k) an ability to use the techniques, skills, and modern engineering tools

	necessary for engineering practice.		
Contribution to Professional Component:	engineering science: 2 credit or 50% engineering design: 2 credits or 50%		
Prepared by:	Ranga R. Vemuri, Ph.D.	**Date:**	

Tentative Schedule and Grading

Assignment	*Start Date*	*Due Date*
Homework 1	Sep 9	Sep 20
Homework 2	Sep 21	Oct 2
Homework 3	Oct 5	Oct 18
Homework 4	Oct 19	Nov 2
Exam 1	Oct 21	
Project (Progress reports' due dates TBD)	Nov 4	Nov 30
Project Demo	Dec 2 (by signup)	
Exam 2	During Exam Week, As Scheduled by the University	

1. Number of homeworks is tentative. Exact homework and project due dates will be decided as we go. (This year, we may drop the project and increase the number of homeworks. We will discuss this in the class.)

2. There will be some announced quizzes and some pop (unannounced) quizzes.

3. Grading will be based on three components of evaluation: quizzes/homeworks, exams and project. These components 35%, 30% and 35% weight respectively. The two exams will carry equal weight. Relative weights among the homeworks/quizzes will be decided later based on the relative size and complexity of the homeworks/quizzes. (This grading policy may change if the project is dropped from the course and the number of homeworks is increased.)

Other

1. We will use the university Canvas system (http://canopy.uc.edu). Please familiarize yourself with that system. You will receive homework and project instructions and other announcements through Canvas. We will use the Canvas system primarily for one way communication, from me to you. If you need to communicate with me, please use regular email. Please log in and set up your personal profile, especially your correct email address. If you don't, you won't receive my emails! Please check your email regularly. Important notices concerning the course, homeworks, project etc. will be sent by email.

2. Audits and sit-ins are NOT allowed without prior permission of the instructor. ***NOTE TO AUDIT STUDENTS***: To pass the course, audit students will have to perform above the class average (ie. average among students taking the class for credit) on both the exams. Audit students should not submit the homeworks and the project.

3. Note that the exam week is part of the academic semester. Homeworks may be due during the exam week and the final exam may be held during the exam week.

Ethics Policy

- Incidents of plagiarism will be reported to the appropriate disciplinary committee. Please become familiar with UC policies concerning academic dishonesty.

  See the college Ethics Policy available at the CEAS Page
  http://ceas.uc.edu/programs_degrees/policy_on_plagiarism.html

- Under no circumstances is it acceptable to post problems or solutions from assignments in this course on third party websites.
- Collaboration with classmates or third parties is strictly prohibited in this course unless explicitly stated as allowed on a given assignment.
- Students are only allowed to work with their assigned lab/group partner(s) for all parts of the lab/project/etc. assignment. Collaboration/discussion among different lab/project groups is prohibited unless explicitly allowed in specific assignments.

Course Delivery Mode in Fall 2026

1. This course is scheduled as an online course, delivered synchronously. Attendance is required.

2. All of the **lectures** on MW 12.20-1.50pm will be completed online in synchronous mode. This means that the class meets entirely online, and instruction will occur “live” via Zoom. Invitations that link to scheduled live online lectures will be posted on the course site in Canvas.

3. You may be asked to read specific study material and/or review pre-recorded lectures before some classes. During the class, quizzes would be conducted based on the pre-assigned material.

4. All of the **lab recitations** and software demos will occur on W 6-8.50pm online in synchronous mode also.

5. Students will complete homeworks and projects using software available on the EECS department computers. This software can be accessed and used remotely by online remote access to these computers or by physical access to the VLSI computer lab in Room 810 Rhodes.

   The current plan is for students to complete the homeworks and projects entirely by remote access. However, in the event of remote access or bandwidth issues, those students that are able to physically come to Rhodes 810 may contact the instructor requesting physical access. Physical use policies will be communicated to the students if and when such use becomes necessary and can be allowed.

   Homeworks/labs and projects in this course require the use of VLSI design software which can be remotely accessed but requires a reliable internet connection with high bandwidth. In addition, while not essential, a large monitor and hand-held mouse are highly desirable to make it easier to use the software. Students must plan to spread their work to avoid last minute software access and server overload issues.

   Students are responsible for completing both the individual and group homework assignments and projects by appropriate planning to avoid last minute remote access reliability and speed issues.

Communication

Instructor's email address is ranga.vemuri@uc.edu. Send an email to contact. An online meeting can be set up if necessary.

Attendance and Participation

1. Attendance is required and may be recorded. There will no make up quizzes or excuses for missed quizzes.

2. Class participation is encouraged. If you are able share your notes or quiz solutions by screen sharing, you may be asked to do so.

3. If you have an unexpected IT issue that prevents your participation in a synchronous online session of the course, please contact UC IT for help in resolving your technical issue (https://www.uc.edu/about/ucit/help.html or 513-556-HELP), and notify me as soon as possible of the reason for your absence. Our plan is to record and post the lectures online but this is subject to Canvas/Kaltura related recording issues and delays.

Class Cancellation

1. If an IT issue delays starting of the class or interrupts the class, please wait online. We will try to resolve the issue and (re)start the class as quickly as possible.

2. If a class needs to be cancelled due to IT or other issues, lecture material will provided online or a make up class will be scheduled as necessary.

3. If class schedule needs to be changed in a short notice, we will try our best to communicate as quickly as possible by email.

4. If the University closes due to inclement weather or other emergency situations, there will be university-wide announcements via email and also to your cell phone number on record through the automatic University emergency text messaging system. Students should notify the University if they change their cell phone number to ensure they will receive these important emergency communications. As soon as possible following a University closure, I will post an announcement on Canvas with instructions for how the material for the day will be covered.

AI Policy

Use of AI tools, including generative AI tools such as ChatGPT, is not allowed for completing any coursework such as quizzes, homeworks, and exams, unless specifically allowed. All submitted work is expected to be produced by the students themselves. Any exception to this policy will be explicitly specified if and when needed.

"Simple Syllabus"

For additional information, visit the "Simple Syllabus" link on the Canvas course page,